

Outcome Image Descriptions

This document explains every image currently present in the project's `output/` folder.

Important Note

The three summary plots were generated by `scripts/embedding_vis.py`.

The Grad-CAM images were generated by `scripts/grad_cam.py`.

At the moment, the Grad-CAM script is running in **demo mode** because the trained classifier weights were not found in `models/efficientnet_b0_food101_best.pt`. That means the Grad-CAM figures are useful for explaining the visualization method, but they should **not** be treated as final scientific evidence of the trained classifier's behavior.

1. `tsne_food101.png`

What it is: A 2D t-SNE projection of the Food-101 class-name embeddings created with the `all-MiniLM-L6-v2` sentence-transformer.

What it shows: Each point corresponds to one food class. Foods with similar semantic meaning appear closer together. The colors group foods into broad categories such as desserts, Asian dishes, seafood, salads, breakfast foods, and western main dishes.

Why it matters: This image helps explain why some food labels are naturally close in embedding space. If two foods appear near each other, the downstream nutrition lookup and refinement stages may treat them as semantically related.

![tsne_food101](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\tsne_food101.png)

2. `similarity_heatmap.png`

What it is: A cosine-similarity heatmap computed from representative Food-101 label embeddings.

What it shows: The matrix compares selected food names against one another. Green cells indicate higher similarity, while red cells indicate lower similarity.

Why it matters: This figure gives a direct explanation of which food names are semantically similar. It is useful for understanding why some predictions or nutrition matches may be confused, especially for related dishes such as pasta variants, soups, or breakfast foods.

![similarity_heatmap](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\similarity_heatmap.png)

3. `confusion_pairs.png`

What it is: A horizontal bar chart showing similarity scores for confusion-prone food pairs.

What it shows: Each bar represents a pair of foods that are likely to be confused by a vision system, such as `ramen` vs `pho`, `pancakes` vs `waffles`, or `sushi` vs `sashimi`.

Why it matters: This is one of the most practical figures in the project because it connects semantic similarity to likely classifier mistakes. The higher the similarity, the more reasonable it is that the CV model may confuse those classes, and the more valuable the VLM refinement stage becomes.

![confusion_pairs](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\confusion_pairs.png)

4. gradcam_burger.png

What it is: A Grad-CAM visualization for the `burger.jpg` evaluation image.

What it shows: The figure has three parts: the original image, the heatmap, and the overlay. Red and yellow regions indicate areas the CNN focused on most strongly.

Why it matters: This visualization demonstrates how Grad-CAM can highlight the image regions that influence a food prediction. In a trained system, we would expect attention around the burger itself rather than irrelevant background regions.

Current limitation: This was generated in demo mode, so it illustrates the method rather than a final trained-model explanation.

![gradcam_burger](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_burger.png)

5. gradcam_cake.png

What it is: A Grad-CAM visualization for the `cake.jpg` evaluation image.

What it shows: It highlights the regions of the cake image that the classifier uses when forming its prediction.

Why it matters: For dessert categories, Grad-CAM can reveal whether the model focuses on meaningful features such as frosting layers, texture, or toppings.

Current limitation: Because the trained weights are missing, this result is a demonstration figure only.

![gradcam_cake](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_cake.png)

6. gradcam_icecream.png

What it is: A Grad-CAM visualization for the `icecream.jpg` evaluation image.

What it shows: The heatmap indicates which visual parts of the dessert were most influential for the CNN prediction.

Why it matters: This kind of figure is useful for checking whether the model is looking at the ice cream itself, toppings, or unrelated background content.

Current limitation: This output is still in demo mode and should not be treated as final evaluation evidence.

![gradcam_icecream](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_icecream.png)

7. gradcam_pancakes.png

What it is: A Grad-CAM visualization for the `pancakes.jpg` evaluation image.

What it shows: It reveals the image zones that most contributed to the classifier's pancake-related decision.

Why it matters: This is useful because breakfast foods such as pancakes, waffles, and French toast can be visually similar. Grad-CAM helps explain which features the CNN considered important.

Current limitation: The figure is a visualization demo until the real classifier checkpoint is added.

![gradcam_pancakes](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_pancakes.png)

8. gradcam_pasta.png

What it is: A Grad-CAM visualization for the `pasta.jpg` evaluation image.

What it shows: The overlay marks the regions that were most important for the pasta prediction.

Why it matters: Pasta-like dishes are often confused with related categories such as spaghetti variants or noodle dishes. This figure demonstrates how the CNN explanation tool can be used to inspect such mistakes.

Current limitation: Generated in demo mode, so it is not yet a trained-model interpretation.

![gradcam_pasta](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_pasta.png)

9. gradcam_pizza.png

What it is: A Grad-CAM visualization for the `pizza.jpg` evaluation image.

What it shows: It displays the original pizza image, the activation heatmap, and a blended overlay showing where the classifier concentrated attention.

Why it matters: Pizza is a visually distinctive class, so this kind of figure can be used to confirm whether the CNN is learning intuitive features such as the pizza surface, crust, and toppings.

Current limitation: This is still a demonstration output because the trained checkpoint is not yet present.

![gradcam_pizza](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_pizza.png)

10. gradcam_ramen.png

What it is: A Grad-CAM visualization for the `ramen.jpg` evaluation image.

What it shows: The heatmap highlights the bowl regions, noodles, toppings, or other parts that contribute to the model output.

Why it matters: This is especially relevant because ramen is one of the foods that can be confused with similar soup or noodle dishes such as pho. It helps motivate the use of the VLM stage for refinement.

Current limitation: This image was created in demo mode, so it should be interpreted as a method illustration.

![gradcam_ramen](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_ramen.png)

11. gradcam_salad.png

What it is: A Grad-CAM visualization for the `salad.jpg` evaluation image.

What it shows: The figure indicates which parts of the salad image were most influential in the CNN's internal decision process.

Why it matters: Salads often contain mixed textures and ingredients, so this kind of explanation can reveal whether the model focuses on the actual food content instead of the plate or background.

Current limitation: This is a demo-mode result due to missing trained weights.

![gradcam_salad](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_salad.png)

12. gradcam_steak.png

What it is: A Grad-CAM visualization for the `steak.jpg` evaluation image.

What it shows: It emphasizes the regions that the classifier relied on most when identifying the steak image.

Why it matters: For meat dishes, Grad-CAM can help determine whether the CNN is focusing on texture, grill marks, plating, or surrounding elements.

Current limitation: This output is illustrative only until the final trained classifier weights are available.

![gradcam_steak](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_steak.png)

13. gradcam_sushi.png

What it is: A Grad-CAM visualization for the `sushi.jpg` evaluation image.

What it shows: It marks the regions that most influenced the CNN when processing the sushi example.

Why it matters: This figure is useful because sushi can be confused with visually related seafood classes such as sashimi. A trained Grad-CAM map would help explain the difference in focus.

Current limitation: This is a demo output because the real classifier checkpoint has not yet been added.

![gradcam_sushi](C:\Users\mahes\OneDrive\Desktop\SEMESTER 02\Deep Learning\GitRepo_Project\Project\food-pipeline\output\gradcam_sushi.png)

Final Summary

The `output/` folder currently contains two kinds of visual artifacts:

1. **Embedding analysis figures** that explain semantic similarity between food labels.
2. **Grad-CAM interpretation figures** that explain how a CNN would focus on image regions during classification.

Together, these outputs support the project story well:

- the embedding plots explain semantic relationships and likely label confusion,
- the Grad-CAM plots explain model attention,
- and both provide motivation for adding the VLM refinement stage on top of the CV pipeline.